CIA - 1

DSA0308 - Natural Language Processing

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Slot - D

1) Here, given information system of a hostipal's clinical notes, the four lines are to be processed:

(L1) - Appointment on 05/03/2025 at 10:30.

(L2) - Call 9876543210 or +91-9123456789.

(L3) - ravi.k@college.edu.in at the date of 5/3/2025 is not valid.

(L4) - Invalid : 5876543210 and 98765432101.

So, the regular expression of parts and their marks are:

Parts	Regular Expression	Marks
a) Date	`\b(0[1-9]|[12][1-6])|[03]([0-9]|[1-12])\b)`	4
b) Email	`\b+\@[1-24]\[A-Z a-z]\[a-9'+a-z]\w-\b`	3
c) Mobile Number	`\b(10[5](6-9)\[5][0-9]+[+91]\[0-9]`	2

So, in the line of 2, the mobile number have a mistake or disjunct-ion that is, actually a mobile number (which is Indian Mobile number) only consists of 10 digits excluding "+91" (which is an optional code for Indian Mobile number), but the "+91 9123456789" number consists of 11 digits, so this is how the mobile number was corrected by using NLP structured pattern.

2) Given, The lexicon contains : reg-noun = {fox, cat, dog};
irreg-sg-noun = {goose, mouse};
irreg-pl-noun = {geese, mice} and the orther plural suffixes.

a) In a finite state of automaton for English nominal inflection - gives the states of the words likes which state it belongs to and starting must be - {fox} ~~{cat, dog}~~ and the final one must be - {dog}. From the labelled transistions like isregular singular noun & isregular plural known are the suffixes
{ goose, geese, mouse, mice.

b) By the use of automaton, we can easily decide whether it is morphene sequences, which are acreptable

(i) dog + -s ⇒ dogs (Irreg-pl-noun).

(ii) mouse + -s ⇒ mice (Irreg-pl-noun)

(iii) geese. ⇒ goose (Irreg-sg-noun)

(iv) fox + -s ⇒ foxes (Irreg-pl-nown)

(v) mice + -s ⇒ mouse (Irreg-sg-noun)

c) Lexical, Intermediate & the surface levels of finite States

for foxes ⇒ fox + es

geese ⇒ goose + es (the spelling will be changed)

and also thirer state is orthographic which is done through

e - insertion:

Finite - state Morphological parsing

Word	lexical level	Intermediate level	Surface level
foxes	fox + N + Pl	fox^s#	foxes
cats	cat + N + Pl	cat^s#	cats
geese	goose + N + Pl	gee s#	geese

rule: $E \rightarrow e / [x, s, z]^n \rightarrow s\#$

Ex: fox + s ⇒ Foxes

cats + s ⇒ cats

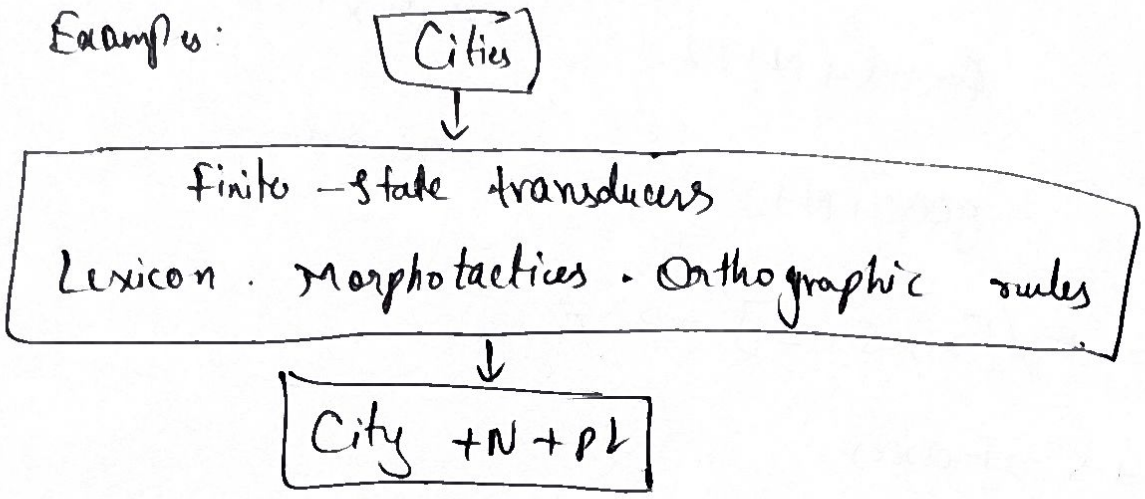
d) Inflectional and Derivational morphology

Word	Type	Reason
Walked	Inflectional	-ed shows past
Unhappiness	Derivational	un-and-ness for new word
Computerised	Derivational	creates a new noun form
Geese	Inflectional	Plural form of geese.

3) (i) The Algorithm is originally originated to apply Porter Stemmer step-by-step in the words called ponies, motoring, relational hopefulness and the important segment is generalizations.

Rule: When the step is used at each stage, have to analyze the stemmer.

Examples:



* Porter Stemmer cascade of rewrite rules

relational → AT, IONAL, ATE → Relate

monitoring → Ing, ed → monitor.

ii) Actually, a training corpus has true sentences: ~~#s~~ #

<s> the cat sata </s>,

<s> the cat ran </s>,

<s> the dog sat </s>,

(S1) is about the The Cat Sat, (S2) The Cat Ran (Past Tense), (S3) The Dog Sat (This is also Past Tense)

4) The three sentences from a small corpus:

<s> I like tea </s>,
<s> I like coffee </s>,
<s> I drink tea </s>.

We can take V = 6 (I, like, tea, coffee, drink; </s>;
<s> is not predicted).

Now N = 12 of training:

a) Counts & MLG:

Unigrams: I = 3, like = 2, tea = 2, coffee = 1,
drink 1, with <s> and </s> = 3 each

Bigrams: (<s>, I) = 3, (I like) = 2, (I drink) = 1
(like, tea) = 1, (like coffee) = 1, (drink tea = 1),
(tea </s> = 2), (coffee </s>) = 1.

b)

5) Part of Speech which is also called as POS by using the Penn Treebank tagset (PRP - proper noun, MD, VB, NN) they must be tag the ambiguous sentences:

Ex: "They can fish"

Here, HMM - Hidden Markov Models parameters are:

$P(PRP) = 0.5$,

$P(NN) = 0.3$,

$P(VB) = 0.1$,

$P(MD) = 0.1$

The transitions of PRP → MD ⇒ 0.4

PRP → NN ⇒ 0.1

~~PRP ⇒~~ MD → VB 0.8

MD → NN 0.2

VB → NN 0.5

⋮

Like that until NN → PRP ⇒ 0.2

This emission: P(they | PRP) = 0.8;

P(can | MD) = 0.3

P(can | VB) = 0.02

P(can | NN) = 0.01

P(fish | NN) = 0.02

P(fish | VB) = 0.01

So, the rule fixes 46 errors and introduces 6 on training data.